AI ETHICS: A REVIEW OF BIAS AND FAIRNESS OF MACHINE LEARNING MODELS

Sonam Sharma
Apex Institute of Technology - CSE
Chandigarh University
Punjab, India
sonam8805@gmail.com

Kamal Singh
Apex Institute of Tecnology-CSE
Chandigarh University
Punjab, India
kamalnegi9548@gmail.com

Piyush Mahawar
Apex Institute of Technology-CSE
Chandigarh University
Punjab, India
piyushmahawar8173@gmail.com

Ayushika Singh Rajput
Apex Institute of Technology-CSE
Chandigarh University
Punjab, India
ayushikarajput1810@gmail.com

Abstract—

Artificial intelligence and machine learning have been vitalized in key areas, such as healthcare, education, finance, and hiring, yet due to its use of data, it may introduce and spread biases. Examples of such systems can exhibit biased output when a misstatement exists in the input material or reflects discrimination in society e.g. a hypothetical recruiting pattern that prefers male candidates as it has been trained on biased past data. The article critically examines AI bias and fairness, such as the process of fostering bias in data, algorithms and judgements, by humans. Among the biases that are discussed in the paper, one may mention sample bias (ineffective sets of data), measurement bias (that presents inaccurate data collection), algorithm bias (prejudiced design choices make general data accurate but unfair to a certain section of society), and societal bias (data reflecting historical injustices).Another type of bias addressed by the paper is that the challenge of determined bias is not simply a technical problem but an ethical and social one that needs an interdisciplinary solution. It compares available tools and solutions, such as the AI Fairness 360 (AIF360) and the Fairlearn toolboxes, which offer us the opportunity to lessen the influence of bias at three stages of the AI life cycle: pre-processing (data adjustment), in-processing (change of the learning process), and post-processing (using the outcomes of the model). This study also shows transparency and in this regard, technologies, such as explainable AI (XAI) can assist the consumer in getting to know the manner in which an AI system came up with the decision. The paper arrives at the conclusion that, despite the significant advances, there are still problems, including assessing small biases, accuracy and fairness, and the necessity to develop new ethical principles and regulations in the context of new technologies like generative AI.

Keywords- Bias, Fairness, Data preprocessing, AI Ethics, Machine Learning, Fairlearn

I. Introduction

Artificial intelligence and machine learning usage is found in a diverse range of critical areas, such as health care, education, finances, hiring, even law. These systems are trained with data, however, when there are errors in the input, or when the input reflects undesirable tendencies in society, the models can present biased results. When past records of hiring biased the hiring in favour of men rather than women, an AI trained on such biased data might still favour male candidates. This is why the question of fairness in AI is so important: it provides that the AI systems do not favor anyone, but rather treat everyone equally and promote no discrimination. A systematic review of this subject reviews all the research available to ascertain the research on how bias arises, and the influence it has on different fields and provide solutions available to it. Scholars have found out that bias could be either data-based, or algorithm-based, or even human judgements in the developmental stages. In order to address this, they suggest using specific measures to remove bias in different stages, like improving data quality, introducing fairness options in the course of training, or altering outcomes after training. They also emphasize the need of the transparency devices, i.e., explainable AI where people could get informed about how AI makes the judgement. Overall, the research reveals that it is not only a technical problem but also an ethical and a social one to address bias and have engineers and policymakers collaborate with society so as to be able to create AI systems that will be fair and reliable and accountable. Some of the industries where the stakes are high have seen the adoption of artificial intelligence (AI) and machine learning (ML) systems roll at a compact rate in decision-making processes, in some cases within healthcare, finance, and criminal justice systems. Such technologies will also lead to the arrival of dramatic changes like increased accuracy in medical diagnoses, improved financial services, and the conversely improved hiring procedures [2]. Nonetheless, the use of AI also brought severe ethical issues and, specifically, responsibility, transparency, and justice [3]. Due to historical and data-oriented trends, these systems run a risk of amplifying and strengthening the pre-existing inequality and creating discriminatory outcomes and the worsening of the perception of the population as they become more embedded in the societal systems [4]. It is a common understanding that bias in the AI systems can be defined as a systematic and unjust error in data or algorithm that systematically outweighs certain groups [5].

In contrast, fairness has an emphasis on the equal treatment of different groups of people, and an equitable outcome [6]. Fairness requires an active and interdisciplinary process that intervenes at all stages of the lifecycle of AI, including the collection of data and initial processing, the training, and deployment of models, instead of addressing specific technical concerns individually [7]. This is an expression of a larger idea that a central tenet of the development of ethical AI is that of justice, alongside duty and transparency [8]. The causal chain between the technical activities to the societal outcomes is a long-standing issue. In the course of data collection, existing social injustices are initially coded into data sets thereby creating skewed training data affecting algorithmic assumptions [9]. Even technically sound models may reproduce discriminatory behaviours if trained on biassed data, perpetuating systemic injustices [10].As an example, although there are no sensitive variables explicitly entered, a credit scoring model has been trained on historical lending information can still discriminate against racial or gender minorities [11]. Since the biased effects are eventually conditioned by interdependent technological, social, and organizational aspects, such interactions draw attention to the pointlessness of considering bias as a distinct computational problem [12]. Bias can be of a number of sources. Lack of diversity in the training datasets can lead to data-related problems, such as sample bias [13], when they do not represent diversity in the real world. Measurement bias is caused by inaccurate and faulty data collection [14]. Archiving past wrongs into the training data lead to the emergence of societal bias and often leads to algorithmic discrimination causing serious moral and legal concerns [15]. Besides data, design options such as optimisation strategies sacrificing minority groups to the greater accuracy of the general population may create an algorithmic bias [16].

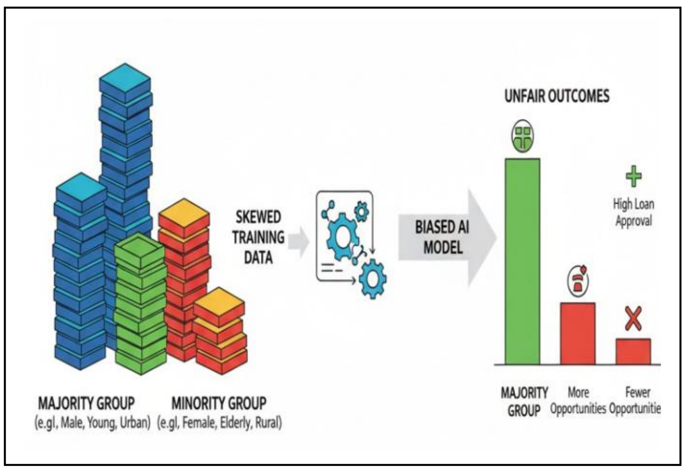

Fig. 1. Data Imbalance Leading to Disparate Outcomes

Objectives

The primary goal of the study is the issues of prejudice and equity have grown due to the rapid application of machine learning (ML) and artificial intelligence (AI) systems to key decision-making domains. Some causes of bias in machine learning models include poor or biased training data, unequal representation of sensitive variables, and algorithmic decisions that reinforce or exacerbate existing injustices in the society accidently. The impacts of such biases are particularly damaging in areas of the criminal justice system, loan issuance, medical diagnoses, and employment processes that involve high stakes and in which discriminatory outcomes may undermine the trust of the community and support systemic discrimination. Although the world has made incredible progress in terms of equalising policies and bias-reduction methods, the field does not contain a unified framework, which would provide both a sense of moral responsibility and technical accuracy. So a systematic examination of the existing strategies and identification of those areas that require greater creativity is necessary. With these challenges in mind, the research will seek to do the following:

To do a thorough research on the causes and symptoms of bias ahead of machine learning pipelines, including feature engineering, data collection, algorithm design, and model implementation.

- In order to assess current measures of fairness, methods and algorithmic interventions impartially, it is preferable to focus on their theory, utility of practices, and drawbacks of that theory.
- We also need to strive to address the higher ethical and social concerns of biased machine learning algorithms and especially in high stakes fields of society like criminal justice, health care, banking, and recruitment.
- To further advance the creation of transparent, fair, and responsible artificial intelligence tools, one needs to determine the unresolved issues and map the possible research directions.

II. Litreature Review

The proper use of AI is referred to as AI ethics, which is used to ensure that systems adhere to such human principles as justice, accountability, and transparency. In this respect, 'Fairness' means ensuring that diverse groups of people are treated and delivered the same outcome in a fair way, and Bias means biased, systematic, and unfair shortcomings in data or algorithms that discriminate against certain groups of people. These concepts are fundamental when deciding on the ethical aspects of machine learning (ML) processes.

Smith et al. have also highlighted the urgency to tackle the ethical and prejudice concerns with the increasing presence of AI and machine learning in clinical and medical practice, and in particular in the area of pathology [1]. Both the harmful and the mitigation opportunities exist at every level of the bias process, which consists of data collection, model creation and human-computer interface. Smith et al. describe AI systems as systems that could be developed and implemented to protect the welfare of patients and ensure fairness [1], and in such a case, the basic ethical principles of autonomy, beneficence, nonmaleficence, and justice must guide an AI system development and implementation. With the help of the FAIR concepts of data management, they propose such strategies as constant monitoring of models, open development approaches, and utilisation of diverse datasets [1].

An Ethical and fair machine learning models structure.

Johnson and Lee [2] provide a complete procedure of bias evaluation and minimization of bias within AI models, including the data preprocessing stage through the deployment of the models. They offer some suggested best practices, such as adversarial debiasing, integrating fairness constraints directly into model architecture, since they recognize that algorithmic and societal bias play a significant role in the real world uses of AI. They also emphasize the importance of effective governance systems and explainable AI (XAI) in helping developers and policymakers to create trustworthy systems. The involvement of multi-ethnic teams and multi-ethnic perspectives among various teams and views are to be upheld by Johnson and Lee [2] in an effort to ensure equity across the AI lifecycle.

The article by Wang et al. [3] is a thorough investigation of the issue of fairness in machine learning considering both the technological and social aspects of the matter. Their review is also based on pre-, in-, and post-processing therapies to ensure less bias. We describe the trade-offs a developer has to take by juxtaposing conventional accuracy metrics with such measures as fairness, such as statistical parity and fair opportunity. Wang et al. [3] explore the long-term fairness sources such as the need to continuous accountability and interdisciplinary collaboration since it highlights the importance of a well-documented accountability, regular testing, as well as stakeholder communication to responsible usage of machine learning.

Approaches, Effects, and Prospects.

Kumar and Patel [4] indicate that developing algorithms should be ethical and with openness and diversification of the datasets and the feature of justice must also be taken into account during the ethical development. They also cover such concerns as intersectionality and unconscious biasness and offer a list of the most developed methods of evaluating and minimizing prejudice. Kumar and Patel [4] claim that the development of the ethical issue should include the aspect of fairness with respect to algorithm creation, diverse datasets, and transparency. They emphasize the importance of the broadleaf strategies, openly availed data, as well as better interpretability as a way of achieving the same outcomes with the representatives of all categories of population.

Brown and Davis [5] reduced one of the instances in which prejudice occurs during the development and use of AI systems that will be used to augment social inequalities. They enter workable solutions that comprise the adherence to algorithm audits quite often, the verification of fairness, and the application of the standardised measuring protocol that can be used to resist the prejudice into practice. Also, as Brown and Davis [5] argues, to carry out a responsible deployment and prevent systemic damage, regulatory systems and interdisciplinary cooperation have to be exercised. To fight bias in dynamic systems, they underline the importance of continuous awareness and the flexibility. Prominent explainable AI models such as LIME and SHAP that enable the creation of transparent and interpretable models are reviewed by Thulasiram [6]. With the aim of enhancing accountability, these technologies can be utilised to reveal model behaviour and help in making moral decision-making. Thulasiram [6] exposes the logic behind the predictions, which makes the use of XAI encourage user trust and assist in revealing hidden biases. He does mention, however, the current lack of scalability and standardization, advising future study to focus on open metrics and cross-disciplinary collaboration in order to create explainable and reliable AI. The authors Patel et al. [7] discuss how the ethical shortcomings of algorithms and their effects on compliance, fairness, and transparency are at risk in AI-based corporate analytics. They explain that bias can be developed in modelling, preprocessing or in data selection that results in unfair results and this destroys confidence and can result in fines by the government. According to Patel et al. [7], such techniques as diversified data sourcing, robust algorithmic auditing, and establishment of governance frameworks in relation to the continuous monitoring are suggested to ensure ethical development and implementation. Furthermore, they recommend the adoption of explainability and transparency throughout the AI lifecycle, alignment of processes with international standards and regulatory requirements, and interdisciplinary collaboration based on the goal of moral business analytics. Focusing on the legal and ethical issues related to the use of AI in credit risk assessment in BNPL e-commerce, Anderson and Moore [8] discuss the issues of fair lending and the likelihood of discrimination. Their study attracts attention to the problems that are caused by unclear machine learning models, inherent biases within credit data, and prediction algorithms that can unintentionally contribute to either promoting unjust lending or discriminating against under-represented populations. In order to fulfill the regulatory standards and responsibility of consumer protection, Anderson and Moore [8] advocate the application of explainable AI tools, rigid compliance inspections, and stricter bias testing. It is also recommended to engage stakeholders and conduct regular audits to take care of the changing nature of bias and prevent reputational and legal issues. The Fairness Eval model introduced by Baraldi et al. [9] provides instructions on how to evaluate and compare the level of fairness of machine learning models in a formal way. The framework allows organisations to select appropriate fairness criteria and determine the model performance based on the factors that are shielded with the help of various datasets such as gender and race. To facilitate the creation of a transparent and fair model in sensitive areas, Baraldi et al. [9] offer the case studies to demonstrate the real-life implementation by detecting and reducing the latent biases. They emphasize the importance of introducing the aspect of fairness assessment in the standard model validation and documentation to encourage proper and ethical use of AI. Lopez et al. [10] tested fairness concerns with machine learning models in audio sentiment analysis, which is a field that is impacted by variability in language, accent, and demographics. Their study questions the use of popular bias reduction measures, indicating that fairness-conscious modelling tools should be used to complement pre-processing and post-processing interventions. The experimental results of Lopez et al. reveal [10] that there is a difference in the sentiment classification performance by user groups. In order to be fair and accountable with audio-based AI systems, they propose to develop representative corpora and serious fairness audits. The more recent developments around the issue of fairness in deep learning systems are gathered together by Parraga et al. [11], where the focus is made on the use of deep learning in Computer Vision and NLP. They discuss the key sources of bias such as training processes, the modelling architecture, and data collection. Parraga et al. [11] consider several of the mitigating measures such as explainable AI methods, trainability, and equity-mindful training. Their effort focuses on the importance of the balance between equity and the model accuracy in the evaluation of fairness. They also recommend using open datasets, community standards and interdisciplinary collaboration to facilitate ethical and fair deep learning processes.

Nadella et al. [12] in their research on the impact of AI on the society identify the importance of governance frameworks in promoting openness, privacy, and equity in the development of AI. To align the AI practices with the social values, their research studies stakeholder participation and participatory governance approaches. Based on effective recommendations, Nadella et al. [12] propose practical recommendations such as establishment of interdisciplinary ethics committees, dynamic regulations, and ethics-by-design models, to help build a sense of trust and social good.Towards this end, Radanliev [12] adds the element of privacy, fairness, and openness as key ethical concepts in AI. To ensure compliance with the moral and legal duties and to retain the confidence of the population, he emphasizes the need to develop in detail, be answerable to algorithms, and conduct periodic reviews of the impacts.

A. *Sources of Bias*

1. *Biased Sources:* Bias in the dataset itself can be caused by limitations of data-related Bias. When training data fails to provide a satisfactory representation of the diversity of the actual users, models generally do not work well with under-represented groups. As an example, speech recognition software that is tailored to the American English language is often incapable of recognizing other accents.
2. *Algorithmic Bias:* Model assumptions, feature selection, and optimization decisions can be used to generate bias. As an example, focusing only on overall correctness may harm minority groups, and including such sensitive aspects as gender or ethnicity may lead to discrimination. Fairness-conscious measures would have to be taken to reduce these risks.
3. *Human and Organizational Bias:* Design of models, use of data and evaluations can be influenced by the decisions and opinions of developers. Organization culture is another reason: when the group values ethics greatly, it is more likely that the practices oriented toward equity are introduced.

B. *Types of Bias*

1. *Sampling bias:* This happens when specific categories of individual or the situation are not represented in the data collected. The model is unjust as some groups may be left out or less attended to.
2. *Measurement bias:* This is the event that occurs when the data is improperly measured or erroneous. The model can be confused with wrong labels, absent values, or poor measurements.
3. *Algorithmic bias:* This bias is an effect of model construction or training. The policies, goals, or techniques used, may benefit one group more than the other.
4. *Societal bias:* This is the consequence of the information being the equalizing disparities in society. As an example, the model can mimic any unfair treatment in case previous discrimination in loans or jobs is included.
5. *Temporal bias:* Temporal bias occurs when models are trained with old data which is no longer accurate or representative of the current trends. The predictions may prove unfair or inaccurate with the change of the circumstances in the course of time.

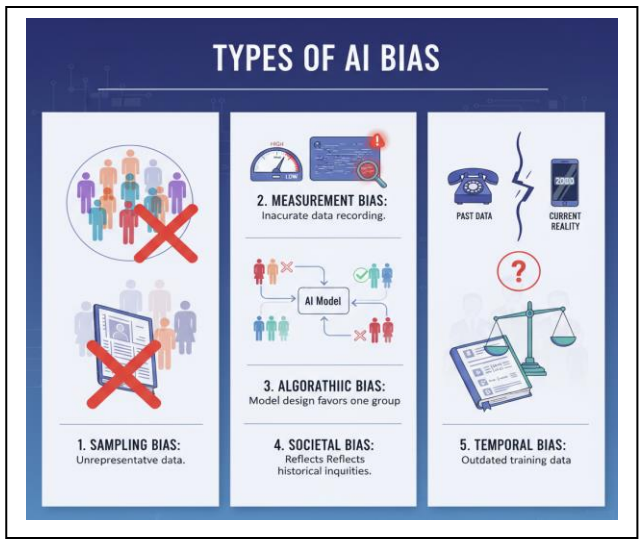

Fig. 2. Types of Bias

C. *Libraries and Tools for Bias Reduction*

1. *AI Fairness 360 (AIF360):* IBM has an open-source package called AI Fairness 360 (AIF360), which provides bias mitigation and fairness options applicable to any part of the AI lifecycle.
2. *Fairlearn:* Python library A library offering algorithms of bias correction and fairness metrics to help find and mitigate fairness.
3. Aequitas: An online bias audit application and toolset, which can generate reports to identify potential systemic biases in a model and data.
4. Bias Metrics and Audits: Two examples of tools providing metrics of fairness to both measure and

audit bias in datasets and model predictions include AIF360 and Fairlearn.

D. Applications

The use of machine learning in various disciplines highlights the importance of going beyond prejudice and being unbiased. The healthcare sector is increasingly implementing AI models to treat patients and diagnose diseases through diagnostic imaging as well as predicting diseases. Nonetheless, the issue of the fairness of the clinical outcomes can be questioned since biased medical data often causes unequal performance in demographic groups. Approaches based on fairness are essential to provide fair treatment of applicants because automated systems that are applied in hiring and recruitment to screen resumes and select the best candidates may reinforce racial and gender inequities in the past. Machine learning is employed in the banking industry to assist in credit rating, defense of fraudulent. It can also be the cause of bias in service refusals using biased datasets that are based on racial or socioeconomic unfair treatments; mitigation methods and measures of fairness can be used so that convinced decisions become more inclusive. Like in this case, AI systems employed by the criminal justice system to handle risks and predictive policing might serve to reinforce biases in criminal information, which might disproportionately affect specific groups of persons. Principles related to fairness should be engaged to ensure justice and also reduce discrimination tendencies.

III. Methodology

A. Flowchart

As far as ensuring that AI models are equitable, there are four key steps.. To begin with, data collection and exploration can facilitate the understanding of the dataset by identifying such issues as overlooked numbers, disproportionate distribution of classes, or unbalanced representation. Second, to find out the disadvantaged indicator of specific groups, the identification of bias is conducted with the assistance of the fairness standards like demographic parity or equal opportunity. Third, mitigation of bias is implemented, such as resampling data, placing fairness constraints, employing algorithms such as Fairlearn that are fairness-centred during model training. As an illustration, to reduce bias prediction, fairness constraints can be incorporated into a random forest classification model. Finally, through the application of outcome monitoring, the deployment of ethical models is ensured through the avoidance of bias drift and the maintenance of the model as fair and accurate in reality by constantly tracking fairness indicators and performance.

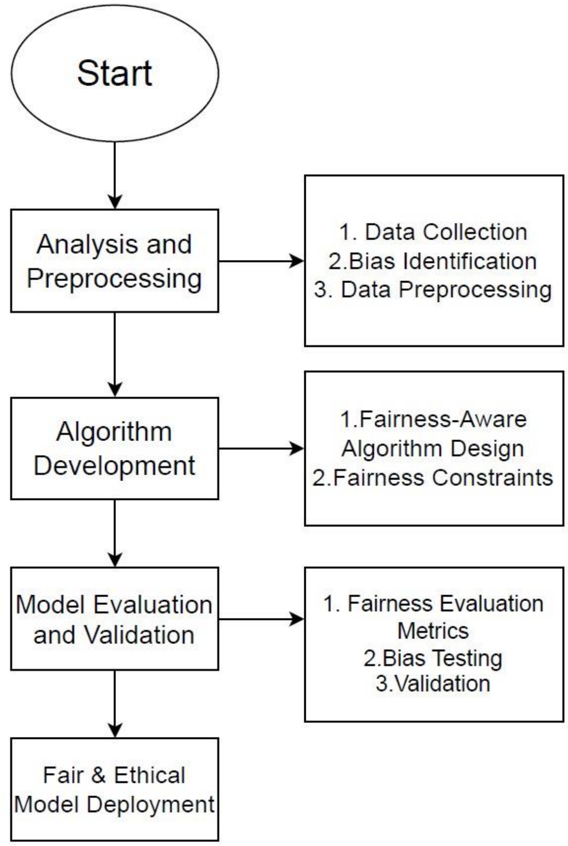

Fig. 3. Flowchart of the Methodology

B. Preparing and processing data

The Prior study points at the need to comprehend datasets prior to using ML models. Data sets are initially collected and researched in order to analyze their structure and find any possible biases. Minorities or women can be underrepresented by the hiring of datasets, e.g. In order to establish the causes of bias, statistical methods including chi-square analysis and distribution analysis are used together with subject matter expertise. To overcome these biases, preprocessing techniques, including SMOTE (Synthetic Minority Oversampling Technique), are commonly employed in order to enhance data and reduce undersampling, as well as balance classes. These steps reduce the risk of bias on the model findings and ensure that the input information is representative.

C. Algorithms Development

Research suggests that fairness must be explicitly embedded as a part of the design of machine learning algorithms. The adaptation of popular models like Random Forest, Gradient Boosting, and logistic regression can include the aspect of fairness. Considering the training of the model, fairness constraints are provided, including equal opportunity (equal chances to be able to make the correct prediction of all groups) and demographic parity (equal predictions of all groups). Moreover, strategies such as adversarial debiasing and fairness regularization can reduce discriminatory behaviour by punishing biased predictions or hiding sensitive features, in order to reduce the effect of discriminatory behaviour in the models.

D. Model Assessment and Model Validation

Besides the accuracy, the study notes that it is important to assess models based on fairness measures. The most common metrics are statistical parity difference, equalized odds and disparate impact ratio. Bias testing is conducted in order to ensure that models do not regularly advantage or hurt any given group. As an illustration, a credit scoring algorithm should fairly approve of both male and female applicants having similar financial profiles.

E. Outcome:

- *ETHICAL AND FAIR OUTCOME MODEL DEPLOYMENT:* It has been suggested by research that models should be used in a manner that ensures fair and high accuracy. As an example, the healthcare diagnosis models should work in the same way with the patients of any age and gender.
- *CONTINUOUS MONITORING:* Continuous monitoring according to the current body of research is essential to monitor fairness over time to detect model drift which is the tendency of a previously fair model to turn biased due to the introduction of new data patterns. Indicatively, with a changing labour market data, recruitment models may have to be periodically revisited, to ensure that no group is unintentionally favoured.

IV. CHALLENGES

There are still many obstacles and restrictions in place despite tremendous advancements in our knowledge of and ability to eliminate bias in machine learning. These difficulties stem from how difficult it is to identify prejudice, strike a balance between performance and justice, and implement solutions in practical settings. Furthermore, there are additional challenges due to social, legal, and regulatory concerns. The main obstacles to guaranteeing ethics and fairness in AI systems are highlighted by the following points:

- *Identifying and Measuring Bias:* Bias in AI is typically subtle and not easy to detect. It may not show in overall results but occurs in smaller subgroups (e.g., women of a given race). Different fairness methods (like demographic parity or equal opportunities) may produce inconsistent results, making it hard to choose the proper one. Poor or inadequate data can also disguise bias, and since social conditions vary over time, fairness must be reviewed routinely.
- *Accuracy vs. Fairness Trade-off:* Accuracy can occasionally suffer when fairness is increased. For instance, guaranteeing equal opportunities for all groups may impair a model's overall performance. Context determines these trade-offs: accuracy may be more important in the healthcare industry, whereas fairness may be more important in the criminal justice system. Ethical decisions are frequently required when choosing between the two.
- *Mitigation Methods' Limitations:* There are disadvantages to bias-correcting techniques:
  - ➢ Data preprocessing might not eliminate unconscious biases.
  - ➢ Modifications made during processing are intricate and model-specific.
  - ➢ Results may be distorted by post-processing adjustments.
  - ➢ Since it is more difficult to gauge fairness in text, photos, or videos, generative AI presents additional difficulties.
- *Regulatory and Legal Difficulties:* Compliance is challenging because AI regulations are still developing and vary by nation. When bias happens, it's also unclear who should bear the blame—the developer, the business, or third parties. Trust difficulties may arise when ethical expectations and legal standards diverge.
- *Social and Ethical Concern:* Fairness is social as well as technical. AI solutions need to take context-specific ethical and cultural norms into account. Progress is slowed when ethicists, legislators, and computer scientists don't work together.

V. CONCLUSION

This analysis demonstrates that, while progress has been achieved in recognising and resolving bias in machine learning models, some substantial ethical and practical difficulties remain to be overcome. Bias in AI can emerge from data, algorithms, or human decisions, and if not controlled appropriately, it can result in inequity and unfair treatment in automated systems. Solutions have been proposed, such as bias-reducing tools and explainable AI, but each has flaws and limitations; such as, we have to deal with improving fairness as well as retaining accuracy. What applies not merely as a technical problem but also as desirable is to guarantee justice through a strong adherence to moral compromises, continuous monitoring, and government, legislative, and academic collaboration. In order to make sure that AI models are satisfied, just, and trustworthy across each domain of society, the future prosperity will be additionally based on developing targeted and scenario-specific approaches to being fair, the expansion of the legal and regulatory frameworks, and the improvement of inter-disciplinary collaboration.

VI. FUTURE SCOPE

The field of AI ethics, revolves around fairness and bias in machine learning, remains nascent. Although researchers have already detected numerous issues and developed tools to mitigate bias, many important areas require further work in the future:

- *Making Fairness Clear and Practical:* Nowadays, there are many different ways to quantify fairness in AI, and no single measure is suitable for every context. Indeed, some fairness rules are in conflict with one another. In the future, researchers should:
    - They might develop principles of applying the rules of fairness depending on specific fields (hiring or healthcare or criminal justice).
    - Allow individual justice, thus, individuals who are quite broadly similar to each other are treated similarly by an AI.

- *Dealing with Real-World Complexities:* AI systems do not operate in a vacuum, they affect people and the society. This creates new challenges:
    - *Intersectional justice:* The unjust conditions can be encountered by the individuals as a consequence of various identities (ex: gender and race). The critical prejudice, the latent strata of prejudice, needs to be identified by the new AI and fixed.
    - *Fairness gerrymandering:* There is a model also that can appear fair on a broader basis but may be disguising unfairness in smaller groups of individuals. The systems in future auditing must also cross-check of all subgroups.

- *With Causality of Fairness:* The bulk of fairness studies nowadays are on a statistical basis, in which we can only see correlations. However, justice frequently must concern causes - such as, as an example, that someone was rejected due to sex.
    - Any responsible future work must also attempt to uncover true sources of unfairness by reliance on causal reasoning.
    - Counterfactual fairness should be enhanced. That is, seeing whether a person would have otherwise gotten the same decision had only their sensitive attribute (sex or skin colour) been different.
- *Fairness in Generative AI (Chatbots, Image Generators, etc.):* New AI systems such as chatbots (LLMs) and image generators produce complex material rather than straightforward predictions, which can exclude groups or propagate stereotypes.
    - Define new types of harms, such as stereotyping, erasure of certain groups, or spreading toxic content.
    - Develop new metrics to check fairness in text, images, and videos.
    - Teach AI to be context-aware: sometimes fairness means treating everyone the same, but other times it means considering differences (e.g., respecting cultural or religious needs).
- *Regulations, Laws, and Auditing:* Lawmakers are now draughting AI-related legislation, such as the EU AI Act. This implies that compliance must be a key component of AI research:
    - Legal regulations are being transformed into technical safeguards that AI systems can abide by.
    - Creating auditing techniques to determine whether AI is truly equitable, especially in cases where businesses withhold information and the system is a "black box".

References

[1] M. G. Hanna et al, "Ethical and bias implications of applicability of AI and machine learning systems," 2025.

[2] S. Kamatala, P. Naayini, P. K. Myakala, A systematic method to reduce bias towards ethical and fair neural networks, 2025.

[3] R. T. Rabonato and L. Berton, "Fairness strategies in machine learning systems: comprehensive review," 2025.

[4] Bias and fairness in deep learning: Methods, effects and prospects, by M. Shah and N. Sureja, Archives of Computational Methods in Engineering, 2025.

[5] P. Pulivarthy and P. Whig, “Bias and fairness: Addressing discrimination in AI systems,” in Ethical Dimensions of AI Development, Hershey, PA: IGI Global, 2025, pages 103–126.

[6] P. P. Thulasiram, Explainable AI: A more transparent and human-trusted model, 2025.

[7] J. K. Bahangolu and L. Owusu-Berko, "Algorithmic bias and governance: ensuring ethical, transparent AI analytics, 2025.

[8] A. Mishra et al., Regulatory and ethical challenges in AI-driven and machine learning credit risk assessment for Buy Now, Pay Later (BNPL) in U.S. e-commerce, 2025.

[9] FairnessEval: Framework Framework to compute fairness in ML models, A. Baraldi et al., Advances in Database Technology-EDBT, 2025.

[10] S. Luitel, Y. Liu, and M. Anwar, “Investigating fairness in machine learning-based audio sentiment analysis,” AI and Ethics, volume 5, issue 2, pages 1099–1108, 2025.

[11] O. Parraga et al., "Fairness in deep perception of language and vision: An overview," 2025.

[12] Sunil et al., 2022, American Clinical and Community Pharmacy, Ethics in AI: Drafting responsible and transparent practices.

[13] P. Radanliev, Making AI systems fair, transparent, and private, 2025.

[14] J. Umar and J. Reuben, Fostering transparency in AI to support real estate and insurance models, 2025.

[15] E. Barbierato et al., The evolving AI and machine learning: Ontological understanding, 2025.

[16] S. Rao and T. Zhao, “Ethical AI in HR: A case study of tech hiring,” Journal of Computer Information Systems, pages 1–18, 2025.

[17] N. Malali, AI ethics in financial services globally: 2025 and modern perspective.

[18] A. Tiwari and H. E. Farag, Responsible AI framework of autonomous vehicles: Fairness and risk of bias management, 2025.

[19] J. Mandela, Bias and fairness analysis in AI systems based on machine learning 2025.

[20] O. R. Odumbo, Explainable AI and federated learning in healthcare supply chain intelligence: Addressing ethical constraints, bias mitigation, and regulatory compliance for global pharmaceutical distribution, 2025.

[21] R. K. Debbadi and O. Boateng, AI-based automation of business, predictive machine learning, and UiPath, 2025.

[22] E. Sahin, N. N. Arslan and D. Ozdemir, Interpretable deep learning: Challenges and reliability, 2025.

[23] H. Allam et al., Bias in AI: A comparative analysis of DeepSeek and ChatGPT, 2025.

[24] A. Tripathi and V. Kumar, Ethical practices of artificial intelligence: A management framework for responsible AI deployment in businesses, 2025.

[25] K. W. Nathim et al., "Finding a trade-off between reducing bias and a fair design of an ML model," 2024.

[26] A. P. Onebunne and B. Alade, “Bias and fairness in AI models: Addressing disparities in machine learning applications,” International Research Journal of Modernization in Engineering Technology and Science, volume 6, issue 9, 2024.

[27] N. Kodakandla, “Scaling AI responsibly: Leveraging MLOps for sustainable machine learning deployments,” International Journal of Science and Research Archive, volume 13, issue 1, pages 3447–3455, 2024.

[28] D. Ueda et al., “Fairness of artificial intelligence in healthcare: Review and recommendations,” Japanese Journal of Radiology, volume 42, issue 1, pages 3–15, 2024.

[29] A. A. R. Gomez and M. L. C. Benavides, “Framework for bias detection in machine learning models: A fairness approach,” in Proc. 17th ACM International Conference on Web Search and Data Mining (WSDM), 2024.

[30] S. Sharma and D. Seth, “Blue monkey updated chimp optimization algorithm to load balance, 2024.

[31] S. Sharma, R. P. Mahapatra, M. Kapil, and D. Seth, Advanced deployment strategies for elastic load balancing in AWS: A comprehensive study on multi-tier architecture optimization, 2024.

[32] S. Sharma, R. P. Mahapatra, M. Kapil, and D. Seth, Machine learning driven load balancing in software-defined networks: Recent progress and emerging challenges, 2024.

[33] S. Sharma, D. Seth, and M. Kapil, “Combined optimization strategy: CUBW for load balancing in software-defined network,” Web Intelligence, 2024.

[34] S. Camarena, Community-led AI sustainability: a study of Victorian farmers markets, Sustainability, 2021.

[35] S. Parsheera, A gendered approach to AI fairness and inclusion, Proc. ITU Kaleidoscope: ML for a 5G Future, 2018.

[36] S. Venkatasubbu and G. Krishnamoorthy, Ethical aspects of bias and fairness in ML, 2022.

[37] B. Giovanola and S. Tiribelli, Beyond bias and discrimination: Redefining the AI ethics principle of fairness in healthcare machine-learning algorithms, 2023.